

$$S''(-\sqrt{2k}) = -\sqrt{2k}\pi \cdot \cos(k\pi) = -\sqrt{2k}\pi \cdot (-1) = \sqrt{2k}\pi > 0.$$

由此得 $S(x)$ 的全体极小值点为

$$x = (-1)^k \sqrt{2k}, \quad k = 1, 2, 3, \dots$$

17. 设 $f(x)$ 在 $[0, +\infty)$ 内连续, 且 $\lim_{x \rightarrow +\infty} f(x) = 1$. 证明函数

$$y = e^{-x} \int_0^x e^t f(t) dt$$

满足方程 $\frac{dy}{dx} + y = f(x)$, 并求 $\lim_{x \rightarrow +\infty} y(x)$.

证
$$\frac{dy}{dx} = -e^{-x} \int_0^x e^t f(t) dt + e^{-x} \cdot e^x f(x) = -y + f(x),$$

因此 $y(x)$ 满足微分方程 $\frac{dy}{dx} + y = f(x)$.

由条件 $\lim_{x \rightarrow +\infty} f(x) = 1$, 从而存在 $X_0 > 0$, 当 $x > X_0$ 时, 有

$$f(x) > \frac{1}{2}.$$

因此,

$$\begin{aligned} \int_0^x e^t f(t) dt &= \int_0^{X_0} e^t f(t) dt + \int_{X_0}^x e^t f(t) dt \\ &\geq \int_0^{X_0} e^t f(t) dt + \int_{X_0}^x \frac{1}{2} e^t dt \\ &= \int_0^{X_0} e^t f(t) dt + \frac{1}{2} e^{X_0} (x - X_0), \end{aligned}$$

故, 当 $x \rightarrow +\infty$ 时, $\int_0^x e^t f(t) dt \rightarrow +\infty$, 从而利用洛必达法则, 有

$$\lim_{x \rightarrow +\infty} y(x) = \lim_{x \rightarrow +\infty} \frac{\int_0^x e^t f(t) dt}{e^x} = \lim_{x \rightarrow +\infty} \frac{e^x f(x)}{e^x} = 1.$$

习题 5-3 定积分的换元法和分部积分法

1. 计算下列定积分:

$$(1) \int_{\frac{\pi}{3}}^{\pi} \sin \left(x + \frac{\pi}{3} \right) dx;$$

$$(2) \int_{-2}^1 \frac{dx}{(11 + 5x)^3};$$

$$(3) \int_0^{\frac{\pi}{2}} \sin \varphi \cos^3 \varphi d\varphi;$$

$$(4) \int_0^{\pi} (1 - \sin^3 \theta) d\theta;$$

$$(5) \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \cos^2 u du;$$

$$(6) \int_0^{\sqrt{2}} \sqrt{2 - x^2} dx;$$

$$(7) \int_{-\sqrt{2}}^{\sqrt{2}} \sqrt{8-2y^2} dy;$$

$$(9) \int_0^a x^2 \sqrt{a^2 - x^2} dx \quad (a > 0);$$

$$(11) \int_{-1}^1 \frac{x dx}{\sqrt{5-4x}};$$

$$(13) \int_{\frac{3}{4}}^1 \frac{dx}{\sqrt{1-x}-1};$$

$$(15) \int_0^1 t e^{-\frac{t^2}{2}} dt;$$

$$(17) \int_{-2}^0 \frac{(x+2) dx}{x^2+2x+2};$$

$$(19) \int_{-\pi}^{\pi} x^4 \sin x dx;$$

$$(21) \int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{(\arcsin x)^2}{\sqrt{1-x^2}} dx;$$

$$(23) \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos x \cos 2x dx;$$

$$(25) \int_0^{\pi} \sqrt{1+\cos 2x} dx;$$

$$(8) \int_{\frac{1}{\sqrt{2}}}^1 \frac{\sqrt{1-x^2}}{x^2} dx;$$

$$(10) \int_1^{\sqrt{3}} \frac{dx}{x^2 \sqrt{1+x^2}};$$

$$(12) \int_1^4 \frac{dx}{1+\sqrt{x}};$$

$$(14) \int_0^{\sqrt{2}a} \frac{x dx}{\sqrt{3a^2-x^2}} \quad (a > 0);$$

$$(16) \int_1^{e^2} \frac{dx}{x \sqrt{1+\ln x}};$$

$$(18) \int_0^2 \frac{x dx}{(x^2-2x+2)^2};$$

$$(20) \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 4 \cos^4 \theta d\theta;$$

$$(22) \int_{-5}^5 \frac{x^3 \sin^2 x}{x^4+2x^2+1} dx;$$

$$(24) \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{\cos x - \cos^3 x} dx;$$

$$(26) \int_0^{2\pi} |\sin(x+1)| dx.$$

解 (1) $\int_{\frac{\pi}{3}}^{\pi} \sin \left(x + \frac{\pi}{3} \right) dx = \int_{\frac{\pi}{3}}^{\pi} \sin \left(x + \frac{\pi}{3} \right) d \left(x + \frac{\pi}{3} \right)$

$$= \left[-\cos \left(x + \frac{\pi}{3} \right) \right]_{\frac{\pi}{3}}^{\pi} = 0.$$

$$(2) \int_{-2}^1 \frac{dx}{(11+5x)^3} = \int_{-2}^1 \frac{d(11+5x)}{5(11+5x)^3} = \left[-\frac{1}{10(11+5x)^2} \right]_{-2}^1 = \frac{51}{512}.$$

$$(3) \int_0^{\frac{\pi}{2}} \sin \varphi \cos^3 \varphi d\varphi = - \int_0^{\frac{\pi}{2}} \cos^3 \varphi d(\cos \varphi) = \left[-\frac{1}{4} \cos^4 \varphi \right]_0^{\frac{\pi}{2}} = \frac{1}{4}.$$

$$(4) \int_0^{\pi} (1 - \sin^3 \theta) d\theta = \pi + \int_0^{\pi} (1 - \cos^2 \theta) d(\cos \theta)$$

$$\stackrel{u = \cos \theta}{=} \pi + \int_1^{-1} (1 - u^2) du = \pi - \frac{4}{3}.$$

$$(5) \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \cos^2 u du = \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (1 + \cos 2u) du$$

$$= \frac{1}{2} \left[u + \frac{1}{2} \sin 2u \right]_{\frac{\pi}{6}}^{\frac{\pi}{2}} = \frac{\pi}{6} - \frac{\sqrt{3}}{8}.$$

$$(6) \int_0^{\sqrt{2}} \sqrt{2-x^2} dx \stackrel{x=\sqrt{2}\sin u}{=} \int_0^{\frac{\pi}{2}} 2\cos^2 u du = 2 \cdot \frac{\pi}{4} = \frac{\pi}{2}.$$

$$\begin{aligned} (7) \int_{-\sqrt{2}}^{\sqrt{2}} \sqrt{8-2y^2} dy &\stackrel{y=2\sin u}{=} \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} 4\sqrt{2}\cos^2 u du \\ &= 2\sqrt{2} \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} (1+\cos 2u) du \\ &= 2\sqrt{2} \left[u + \frac{1}{2} \sin 2u \right]_{-\frac{\pi}{4}}^{\frac{\pi}{4}} = \sqrt{2}(\pi+2). \end{aligned}$$

$$\begin{aligned} (8) \int_{\frac{1}{\sqrt{2}}}^1 \frac{\sqrt{1-x^2}}{x^2} dx &\stackrel{x=\sin u}{=} \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{\cos^2 u}{\sin^2 u} du = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (\csc^2 u - 1) du \\ &= [-\cot u - u]_{\frac{\pi}{4}}^{\frac{\pi}{2}} = 1 - \frac{\pi}{4}. \end{aligned}$$

$$\begin{aligned} (9) \int_0^a x^2 \sqrt{a^2-x^2} dx &\stackrel{x=a\sin u}{=} \int_0^{\frac{\pi}{2}} a^4 \sin^2 u \cos^2 u du = \frac{a^4}{8} \int_0^{\frac{\pi}{2}} (\sin 2u)^2 d(2u) \\ &\stackrel{t=2u}{=} \frac{a^4}{8} \int_0^{\pi} \sin^2 t dt = \frac{a^4}{4} \int_0^{\frac{\pi}{2}} \sin^2 t dt \\ &= \frac{a^4}{4} \cdot \frac{\pi}{4} = \frac{\pi}{16} a^4. \end{aligned}$$

$$(10) \int_1^{\sqrt{3}} \frac{dx}{x^2 \sqrt{1+x^2}} \stackrel{x=\frac{1}{u}}{=} \int_1^{\frac{1}{\sqrt{3}}} \frac{-u}{\sqrt{1+u^2}} du = [-\sqrt{1+u^2}]_1^{\frac{1}{\sqrt{3}}} = \sqrt{2} - \frac{2\sqrt{3}}{3}.$$

$$(11) \text{ 令 } u = \sqrt{5-4x}, \text{ 即 } x = \frac{5-u^2}{4}, \text{ 得}$$

$$\int_{-1}^1 \frac{x dx}{\sqrt{5-4x}} = \int_3^1 \frac{u^2-5}{8} du = \left[\frac{u^3}{24} - \frac{5}{8}u \right]_3^1 = \frac{1}{6}.$$

$$(12) \text{ 令 } u = \sqrt{x}, \text{ 即 } x = u^2, \text{ 得}$$

$$\int_1^4 \frac{dx}{1+\sqrt{x}} = \int_1^2 \frac{2udu}{1+u} = [2u - 2\ln(1+u)]_1^2 = 2 + 2\ln \frac{2}{3}.$$

$$(13) \text{ 令 } u = \sqrt{1-x}, \text{ 即 } x = 1-u^2, \text{ 得}$$

$$\int_{\frac{3}{4}}^1 \frac{dx}{\sqrt{1-x}-1} = \int_{\frac{1}{2}}^0 \frac{-2udu}{u-1} = -2[u + \ln(1-u)]_{\frac{1}{2}}^0 = 1 - 2\ln 2.$$

$$(14) \int_0^{\sqrt{2}a} \frac{x dx}{\sqrt{3a^2-x^2}} = -\frac{1}{2} \int_0^{\sqrt{2}a} \frac{d(3a^2-x^2)}{\sqrt{3a^2-x^2}}$$

$$= - [\sqrt{3a^2 - x^2}]_0^{\sqrt{2}a} = (\sqrt{3} - 1)a.$$

$$(15) \int_0^1 t e^{-\frac{t^2}{2}} dt = - \int_0^1 e^{-\frac{t^2}{2}} d\left(-\frac{t^2}{2}\right) = \left[-e^{-\frac{t^2}{2}}\right]_0^1 = 1 - e^{-\frac{1}{2}}.$$

$$(16) \int_1^{e^2} \frac{dx}{x\sqrt{1+\ln x}} \stackrel{x=e^u}{=} \int_0^2 \frac{du}{\sqrt{1+u}} = [2\sqrt{1+u}]_0^2 = 2\sqrt{3} - 2.$$

$$\begin{aligned} (17) \int_{-2}^0 \frac{(x+2)dx}{x^2+2x+2} &= \int_{-2}^0 \frac{(x+1)+1}{(x+1)^2+1} dx \\ &= \left[\frac{1}{2} \ln(x^2+2x+2) + \arctan(x+1) \right]_{-2}^0 \\ &= \frac{\pi}{2}. \end{aligned}$$

(18) 令 $x = 1 + \tan u$, 则 $dx = \sec^2 u du$, 因此

$$\begin{aligned} \int_0^2 \frac{x dx}{(x^2 - 2x + 2)^2} &= \int_0^2 \frac{x dx}{[(x-1)^2 + 1]^2} = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{(1 + \tan u) du}{\sec^2 u} \\ &= 2 \int_0^{\frac{\pi}{4}} \cos^2 u du = \int_0^{\frac{\pi}{4}} (1 + \cos 2u) du \\ &= \frac{\pi}{4} + \frac{1}{2}. \end{aligned}$$

(19) 由于被积函数为奇函数, 因此 $\int_{-\pi}^{\pi} x^4 \sin x dx = 0$.

(20) 由于被积函数为偶函数, 因此

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 4 \cos^4 \theta d\theta = 2 \int_0^{\frac{\pi}{2}} 4 \cos^4 \theta d\theta = 8 \cdot \frac{3}{4} \cdot \frac{\pi}{4} = \frac{3}{2} \pi.$$

(21) 由于被积函数为偶函数, 因此有

$$\begin{aligned} \int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{(\arcsin x)^2}{\sqrt{1-x^2}} dx &= 2 \int_0^{\frac{1}{2}} \frac{(\arcsin x)^2}{\sqrt{1-x^2}} dx \\ &= 2 \int_0^{\frac{1}{2}} (\arcsin x)^2 d(\arcsin x) \\ &= \frac{2}{3} [(\arcsin x)^3]_0^{\frac{1}{2}} = \frac{\pi^3}{324}. \end{aligned}$$

(22) 由于被积函数为奇函数, 因此

$$\int_{-5}^5 \frac{x^3 \sin^2 x}{x^4 + 2x^2 + 1} dx = 0.$$

$$(23) \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos x \cos 2x dx = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos x (1 - 2\sin^2 x) dx$$

$$\begin{aligned}
 &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (1 - 2\sin^2 x) d(\sin x) \\
 &= \left[\sin x - \frac{2}{3} \sin^3 x \right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = \frac{2}{3}.
 \end{aligned}$$

或者

$$\begin{aligned}
 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos x \cos 2x dx &= \frac{1}{2} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (\cos 3x + \cos x) dx \\
 &= \frac{1}{2} \left[\frac{1}{3} \sin 3x + \sin x \right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = \frac{2}{3}.
 \end{aligned}$$

$$\begin{aligned}
 (24) \quad \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{\cos x - \cos^3 x} dx &= 2 \int_0^{\frac{\pi}{2}} \sqrt{\cos x} \sin x dx \\
 &\stackrel{u = \cos x}{=} -2 \int_1^0 \sqrt{u} du = \frac{4}{3}.
 \end{aligned}$$

$$(25) \quad \int_0^{\pi} \sqrt{1 + \cos 2x} dx = \int_0^{\pi} \sqrt{2} |\cos x| dx = 2\sqrt{2}.$$

$$(26) \quad \int_0^{2\pi} |\sin(x+1)| dx \stackrel{x=u-1}{=} \int_1^{2\pi+1} |\sin u| du,$$

由于 $|\sin x|$ 是以 π 为周期的周期函数, 因此

$$\text{上式} = 2 \int_0^{\pi} |\sin u| du = 4.$$

2. 设 $f(x)$ 在 $[a, b]$ 上连续, 证明

$$\int_a^b f(x) dx = \int_a^b f(a+b-x) dx.$$

证 令 $x = a + b - u$, 则

$$\begin{aligned}
 \int_a^b f(x) dx &= - \int_b^a f(a+b-u) du = \int_a^b f(a+b-u) du \\
 &= \int_a^b f(a+b-x) dx.
 \end{aligned}$$

3. 证明: $\int_x^1 \frac{dt}{1+t^2} = \int_{\frac{1}{x}}^1 \frac{dt}{1+t^2} \quad (x > 0).$

$$\text{证} \quad \int_x^1 \frac{dt}{1+t^2} \stackrel{t=\frac{1}{u}}{=} - \int_{\frac{1}{x}}^1 \frac{du}{1+u^2} = \int_1^{\frac{1}{x}} \frac{du}{1+u^2} = \int_{\frac{1}{x}}^1 \frac{dt}{1+t^2}.$$

4. 证明: $\int_0^1 x^m (1-x)^n dx = \int_0^1 x^n (1-x)^m dx \quad (m, n \in \mathbf{N}).$

证 令 $x = 1 - u$, 则

$$\int_0^1 x^m (1-x)^n dx = \int_1^0 (1-u)^m u^n du = \int_0^1 x^n (1-x)^m dx.$$

5. 设 $f(x)$ 在 $[0, 1]$ 上连续, $n \in \mathbf{Z}$, 证明

$$\int_{\frac{n}{2}\pi}^{\frac{n+1}{2}\pi} f(|\sin x|) dx = \int_{\frac{n}{2}\pi}^{\frac{n+1}{2}\pi} f(|\cos x|) dx = \int_0^{\frac{\pi}{2}} f(\sin x) dx.$$

证 令 $x = u + \frac{n}{2}\pi$, 则 $dx = du$, 因此

$$\int_{\frac{n}{2}\pi}^{\frac{n+1}{2}\pi} f(|\sin x|) dx = \int_0^{\frac{\pi}{2}} f\left(\left|\sin\left(u + \frac{n}{2}\pi\right)\right|\right) du$$

$$= \begin{cases} \int_0^{\frac{\pi}{2}} f(\sin u) du, & n \text{ 为偶数,} \\ \int_0^{\frac{\pi}{2}} f(\cos u) du, & n \text{ 为奇数.} \end{cases}$$

$$\int_{\frac{n}{2}\pi}^{\frac{n+1}{2}\pi} f(|\cos x|) dx = \int_0^{\frac{\pi}{2}} f\left(\left|\cos\left(u + \frac{n}{2}\pi\right)\right|\right) du$$

$$= \begin{cases} \int_0^{\frac{\pi}{2}} f(\cos u) du, & n \text{ 为偶数,} \\ \int_0^{\frac{\pi}{2}} f(\sin u) du, & n \text{ 为奇数.} \end{cases}$$

由于 $\int_0^{\frac{\pi}{2}} f(\sin x) dx = \int_0^{\frac{\pi}{2}} f(\cos x) dx$, 因此结论成立.

6. 若 $f(t)$ 是连续的奇函数, 证明 $\int_0^x f(t) dt$ 是偶函数; 若 $f(t)$ 是连续的偶函数, 证明

$\int_0^x f(t) dt$ 是奇函数.

证 记 $F(x) = \int_0^x f(t) dt$, 则有

$$F(-x) = \int_0^{-x} f(t) dt \stackrel{t=-u}{=} - \int_0^x f(-u) du,$$

当 $f(t)$ 为奇函数时, $F(-x) = \int_0^x f(u) du = F(x)$, 故 $\int_0^x f(t) dt$ 是偶函数.

当 $f(t)$ 为偶函数时, $F(-x) = - \int_0^x f(u) du = -F(x)$, 故 $\int_0^x f(t) dt$ 是奇函数.

7. 设 $x = \varphi(y)$ 是单调函数 $y = xe^{x^2}$ 的反函数, 求 $\int_0^e \varphi(y) dy$.

解 记 $f(x) = xe^{x^2}$.

作换元 $y = f(x)$. 当 $y = 0$ 时, $x = 0$; 当 $y = e$ 时, $x = 1$. 于是

$$\begin{aligned}
 \int_0^e \varphi(y) dy &= \int_0^1 \varphi(f(x)) d(f(x)) \\
 &= \int_0^1 x d(f(x)) \\
 &= [xf(x)]_0^1 - \int_0^1 f(x) dx = f(1) - \int_0^1 xe^{x^2} dx \\
 &= e - \frac{1}{2} \int_0^1 e^{x^2} d(x^2) = e - \frac{1}{2} [e^{x^2}]_0^1 \\
 &= e - \frac{1}{2}(e - 1) = \frac{1}{2}(e + 1).
 \end{aligned}$$

8. 计算下列定积分:

- | | |
|--|--|
| (1) $\int_0^1 xe^{-x} dx;$ | (2) $\int_1^e x \ln x dx;$ |
| (3) $\int_0^{\frac{2\pi}{\omega}} t \sin \omega t dt$ (ω 为常数); | (4) $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{x}{\sin^2 x} dx;$ |
| (5) $\int_1^4 \frac{\ln x}{\sqrt{x}} dx;$ | (6) $\int_0^1 x \arctan x dx;$ |
| (7) $\int_0^{\frac{\pi}{2}} e^{2x} \cos x dx;$ | (8) $\int_1^2 x \log_2 x dx;$ |
| (9) $\int_0^{\pi} (x \sin x)^2 dx;$ | (10) $\int_1^e \sin(\ln x) dx;$ |
| (11) $\int_{\frac{1}{e}}^e \ln x dx;$ | (12) $\int_0^1 (1-x^2)^{\frac{m}{2}} dx$ ($m \in \mathbf{N}_+$); |
| (13) $J_m = \int_0^{\pi} x \sin^m x dx$ ($m \in \mathbf{N}_+$). | |

解 (1) $\int_0^1 xe^{-x} dx = - \int_0^1 x d(e^{-x}) = - [xe^{-x}]_0^1 + \int_0^1 e^{-x} dx$

$$= -e^{-1} + [-e^{-x}]_0^1 = 1 - \frac{2}{e}.$$

$$(2) \int_1^e x \ln x dx = \int_1^e \frac{\ln x}{2} d(x^2) = \left[\frac{1}{2} x^2 \ln x \right]_1^e - \int_1^e \frac{x}{2} dx = \frac{e^2 + 1}{4}.$$

$$\begin{aligned}
 (3) \int_0^{\frac{2\pi}{\omega}} t \sin \omega t dt &= - \frac{1}{\omega} \int_0^{\frac{2\pi}{\omega}} t d(\cos \omega t) = - \frac{1}{\omega} [t \cos \omega t]_0^{\frac{2\pi}{\omega}} + \frac{1}{\omega} \int_0^{\frac{2\pi}{\omega}} \cos \omega t dt \\
 &= - \frac{2\pi}{\omega^2} + \frac{1}{\omega^2} [\sin \omega t]_0^{\frac{2\pi}{\omega}} = - \frac{2\pi}{\omega^2}.
 \end{aligned}$$

$$(4) \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{x}{\sin^2 x} dx = - \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} x d(\cot x) = [-x \cot x]_{\frac{\pi}{4}}^{\frac{\pi}{3}} + \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \cot x dx$$

$$\begin{aligned}
&= -\frac{\pi}{3\sqrt{3}} + \frac{\pi}{4} + [\ln \sin x]_{\frac{\pi}{4}}^{\frac{\pi}{3}} \\
&= \left(\frac{1}{4} - \frac{\sqrt{3}}{9} \right) \pi + \frac{1}{2} \ln \frac{3}{2}.
\end{aligned}$$

$$\begin{aligned}
(5) \quad \int_1^4 \frac{\ln x}{\sqrt{x}} dx &= \int_1^4 2 \ln x d\sqrt{x} = [2\sqrt{x} \ln x]_1^4 - \int_1^4 \frac{2}{\sqrt{x}} dx \\
&= 8 \ln 2 - [4\sqrt{x}]_1^4 = 4(2 \ln 2 - 1).
\end{aligned}$$

$$\begin{aligned}
(6) \quad \int_0^1 x \arctan x dx &= \frac{1}{2} \int_0^1 \arctan x d(x^2) \\
&= \left[\frac{1}{2} x^2 \arctan x \right]_0^1 - \frac{1}{2} \int_0^1 \frac{x^2}{1+x^2} dx \\
&= \frac{\pi}{8} - \frac{1}{2} [x - \arctan x]_0^1 = \frac{\pi}{4} - \frac{1}{2}.
\end{aligned}$$

$$\begin{aligned}
(7) \quad \int_0^{\frac{\pi}{2}} e^{2x} \cos x dx &= \frac{1}{2} \int_0^{\frac{\pi}{2}} \cos x d(e^{2x}) \\
&= \frac{1}{2} [e^{2x} \cos x]_0^{\frac{\pi}{2}} + \frac{1}{2} \int_0^{\frac{\pi}{2}} e^{2x} \sin x dx \\
&= -\frac{1}{2} + \frac{1}{4} \int_0^{\frac{\pi}{2}} \sin x d(e^{2x}) \\
&= -\frac{1}{2} + \frac{1}{4} [e^{2x} \sin x]_0^{\frac{\pi}{2}} - \frac{1}{4} \int_0^{\frac{\pi}{2}} e^{2x} \cos x dx,
\end{aligned}$$

因此有

$$\int_0^{\frac{\pi}{2}} e^{2x} \cos x dx = \frac{1}{5} (e^{\pi} - 2).$$

$$\begin{aligned}
(8) \quad \int_1^2 x \log_2 x dx &= \frac{1}{2} \int_1^2 \log_2 x d(x^2) \\
&= \frac{1}{2} [x^2 \log_2 x]_1^2 - \frac{1}{2} \int_1^2 \frac{x}{\ln 2} dx \\
&= 2 - \frac{1}{4 \ln 2} [x^2]_1^2 = 2 - \frac{3}{4 \ln 2}.
\end{aligned}$$

$$\begin{aligned}
(9) \quad \int_0^{\pi} (x \sin x)^2 dx &= \frac{1}{2} \int_0^{\pi} x^2 (1 - \cos 2x) dx \\
&= \frac{\pi^3}{6} - \frac{1}{4} \int_0^{\pi} x^2 d(\sin 2x) \\
&= \frac{\pi^3}{6} - \frac{1}{4} [x^2 \sin 2x]_0^{\pi} + \frac{1}{2} \int_0^{\pi} x \sin 2x dx
\end{aligned}$$

$$\begin{aligned}
&= \frac{\pi^3}{6} - \frac{1}{4} \int_0^\pi x d(\cos 2x) \\
&= \frac{\pi^3}{6} - \frac{1}{4} [x \cos 2x]_0^\pi + \frac{1}{4} \int_0^\pi \cos 2x dx \\
&= \frac{\pi^3}{6} - \frac{\pi}{4}.
\end{aligned}$$

$$\begin{aligned}
(10) \quad \int_1^e \sin(\ln x) dx &\stackrel{x=e^u}{=} \int_0^1 e^u \sin u du = [e^u \sin u]_0^1 - \int_0^1 e^u \cos u du \\
&= e \sin 1 - [e^u \cos u]_0^1 - \int_0^1 e^u \sin u du \\
&= e(\sin 1 - \cos 1) + 1 - \int_0^1 e^u \sin u du,
\end{aligned}$$

$$\text{所以, } \int_1^e \sin(\ln x) dx = \frac{e}{2}(\sin 1 - \cos 1) + \frac{1}{2}.$$

$$\begin{aligned}
(11) \quad \int_{\frac{1}{e}}^e |\ln x| dx &= - \int_{\frac{1}{e}}^1 \ln x dx + \int_1^e \ln x dx \\
&= - [x \ln x]_{\frac{1}{e}}^1 + \int_{\frac{1}{e}}^1 dx + [x \ln x]_1^e - \int_1^e dx \\
&= 2 - \frac{2}{e}.
\end{aligned}$$

$$\begin{aligned}
(12) \quad \int_0^1 (1-x^2)^{\frac{m}{2}} dx &\stackrel{x=\sin u}{=} \int_0^{\frac{\pi}{2}} \cos^{m+1} u du \\
&= \begin{cases} \frac{m}{m+1} \cdot \frac{m-2}{m-1} \cdot \cdots \cdot \frac{1}{2} \cdot \frac{\pi}{2}, & m \text{ 为奇数,} \\ \frac{m}{m+1} \cdot \frac{m-2}{m-1} \cdot \cdots \cdot \frac{2}{3}, & m \text{ 为偶数,} \end{cases} \\
&= \begin{cases} \frac{1 \cdot 3 \cdot 5 \cdot \cdots \cdot m}{2 \cdot 4 \cdot 6 \cdot \cdots \cdot (m+1)} \cdot \frac{\pi}{2}, & m \text{ 为奇数,} \\ \frac{2 \cdot 4 \cdot 6 \cdot \cdots \cdot m}{1 \cdot 3 \cdot 5 \cdot \cdots \cdot (m+1)}, & m \text{ 为偶数.} \end{cases}
\end{aligned}$$

(13) 由教材本节的例 6, 可得

$$J_m = \int_0^\pi x \sin^m x dx = \frac{\pi}{2} \int_0^\pi \sin^m x dx.$$

而

$$\int_0^\pi \sin^m x dx \stackrel{x=\frac{\pi}{2}+t}{=} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^m t dt = 2 \int_0^{\frac{\pi}{2}} \cos^m t dt = 2 \int_0^{\frac{\pi}{2}} \sin^m x dx,$$

故

$$J_m = \pi \int_0^{\frac{\pi}{2}} \sin^m x dx.$$

从而有

$$J_m = \begin{cases} \frac{2 \cdot 4 \cdot 6 \cdots (m-1)}{1 \cdot 3 \cdot 5 \cdots m} \cdot \pi, & m \text{ 为大于 } 1 \text{ 的奇数,} \\ \frac{1 \cdot 3 \cdot 5 \cdots (m-1)}{2 \cdot 4 \cdot 6 \cdots m} \cdot \frac{\pi^2}{2}, & m \text{ 为偶数,} \end{cases}$$

$$J_1 = \pi.$$

习题 5-4 反常积分

 1. 判定下列各反常积分的收敛性,如果收敛,计算反常积分的值:

$$(1) \int_1^{+\infty} \frac{dx}{x^4};$$

$$(2) \int_1^{+\infty} \frac{dx}{\sqrt{x}};$$

$$(3) \int_0^{+\infty} e^{-ax} dx \quad (a > 0);$$

$$(4) \int_0^{+\infty} \frac{dx}{(1+x)(1+x^2)};$$

$$(5) \int_0^{+\infty} e^{-pt} \sin \omega t dt \quad (p > 0, \omega > 0);$$

$$(6) \int_{-\infty}^{+\infty} \frac{dx}{x^2 + 2x + 2};$$

$$(7) \int_0^1 \frac{x dx}{\sqrt{1-x^2}};$$

$$(8) \int_0^2 \frac{dx}{(1-x)^2};$$

$$(9) \int_1^2 \frac{x dx}{\sqrt{x-1}};$$

$$(10) \int_1^e \frac{dx}{x \sqrt{1-(\ln x)^2}}.$$

解 (1) $\int_1^{+\infty} \frac{dx}{x^4} = \left[-\frac{1}{3x^3} \right]_1^{+\infty} = \frac{1}{3}.$

(2) $\int_1^t \frac{dx}{\sqrt{x}} = [2\sqrt{x}]_1^t = 2\sqrt{t} - 2$, 当 $t \rightarrow +\infty$ 时, 该极限不存在, 故该反常积分

发散.

(3) $\int_0^{+\infty} e^{-ax} dx = \left[-\frac{1}{a} e^{-ax} \right]_0^{+\infty} = \frac{1}{a}.$

$$\begin{aligned} (4) \int_0^{+\infty} \frac{dx}{(1+x)(1+x^2)} &= \int_0^{+\infty} \frac{1}{2} \left(\frac{1}{1+x} + \frac{1-x}{1+x^2} \right) dx \\ &= \left[\frac{1}{4} \ln \frac{(1+x)^2}{1+x^2} + \frac{1}{2} \arctan x \right]_0^{+\infty} = \frac{\pi}{4}. \end{aligned}$$

$$\begin{aligned} (5) \int e^{-pt} \sin \omega t dt &= -\frac{1}{p} \int \sin \omega t d(e^{-pt}) \\ &= -\frac{1}{p} e^{-pt} \sin \omega t + \frac{\omega}{p} \int e^{-pt} \cos \omega t dt \end{aligned}$$

$$\begin{aligned}
 &= -\frac{1}{p}e^{-pt}\sin \omega t - \frac{\omega}{p^2}\int \cos \omega t d(e^{-pt}) \\
 &= -\frac{1}{p}e^{-pt}\sin \omega t - \frac{\omega}{p^2}e^{-pt}\cos \omega t - \frac{\omega^2}{p^2}\int e^{-pt}\sin \omega t dt,
 \end{aligned}$$

因此,

$$\int e^{-pt}\sin \omega t dt = \frac{-pe^{-pt}\sin \omega t - \omega e^{-pt}\cos \omega t}{p^2 + \omega^2} + C,$$

故

$$\int_0^{+\infty} e^{-pt}\sin \omega t dt = \left[\frac{-pe^{-pt}\sin \omega t - \omega e^{-pt}\cos \omega t}{p^2 + \omega^2} \right]_0^{+\infty} = \frac{\omega}{p^2 + \omega^2}.$$

$$\begin{aligned}
 (6) \quad \int_{-\infty}^{+\infty} \frac{dx}{x^2 + 2x + 2} &= \int_{-\infty}^0 \frac{d(x+1)}{(x+1)^2 + 1} + \int_0^{+\infty} \frac{d(x+1)}{(x+1)^2 + 1} \\
 &= [\arctan(x+1)]_{-\infty}^0 + [\arctan(x+1)]_0^{+\infty} = \pi.
 \end{aligned}$$

$$(7) \quad \int_0^1 \frac{x dx}{\sqrt{1-x^2}} = [-\sqrt{1-x^2}]_0^1 = 1.$$

$$(8) \quad \int_0^t \frac{dx}{(1-x)^2} = \left[\frac{1}{1-x} \right]_0^t = \frac{1}{1-t} - 1, \text{ 当 } t \rightarrow 1 \text{ 时极限不存在, 故原反常积分}$$

发散.

$$(9) \quad \int_1^2 \frac{x dx}{\sqrt{x-1}} \stackrel{x=u^2+1}{=} 2 \int_0^1 (u^2+1) du = \frac{8}{3}.$$

$$(10) \quad \int_1^e \frac{dx}{x\sqrt{1-(\ln x)^2}} = \int_1^e \frac{d(\ln x)}{\sqrt{1-(\ln x)^2}} = [\arcsin \ln x]_1^e = \frac{\pi}{2}.$$

2. 求由曲线 $y = \frac{1}{4x^2 - 1}$, x 轴和直线 $x = 1$ 所围的向右无限延伸的图形的面积.

解 记此面积为 A , 则有

$$\begin{aligned}
 A &= \int_1^{+\infty} \frac{1}{4x^2 - 1} dx = \frac{1}{2} \int_1^{+\infty} \left(\frac{1}{2x-1} - \frac{1}{2x+1} \right) dx \\
 &= \frac{1}{4} \left[\ln \frac{2x-1}{2x+1} \right]_1^{+\infty} \\
 &= \frac{1}{4} \left(0 - \ln \frac{1}{3} \right) = \frac{1}{4} \ln 3.
 \end{aligned}$$

3. 当 k 为何值时, 反常积分 $\int_2^{+\infty} \frac{dx}{x(\ln x)^k}$ 收敛? 当 k 为何值时, 该反常积分发散? 又

当 k 为何值时, 该反常积分取得最小值?

$$\text{解 } \int \frac{dx}{x(\ln x)^k} = \int \frac{d(\ln x)}{(\ln x)^k} = \begin{cases} \ln \ln x + C, & k=1, \\ -\frac{1}{(k-1)\ln^{k-1} x} + C, & k \neq 1, \end{cases}$$

因此当 $k \leq 1$ 时, 反常积分发散; 当 $k > 1$ 时, 该反常积分收敛, 此时

$$\int_2^{+\infty} \frac{dx}{x(\ln x)^k} = \left[-\frac{1}{(k-1)\ln^{k-1} x} \right]_2^{+\infty} = \frac{1}{(k-1)(\ln 2)^{k-1}}.$$

记 $f(k) = \frac{1}{(k-1)(\ln 2)^{k-1}}$, 则

$$\begin{aligned} f'(k) &= -\frac{1}{(k-1)^2(\ln 2)^{2k-2}} [(\ln 2)^{k-1} + (k-1)(\ln 2)^{k-1} \ln \ln 2] \\ &= -\frac{1 + (k-1) \ln \ln 2}{(k-1)^2(\ln 2)^{k-1}}, \end{aligned}$$

令 $f'(k) = 0$, 得 $k = 1 - \frac{1}{\ln \ln 2}$. 当 $1 < k < 1 - \frac{1}{\ln \ln 2}$ 时, $f'(k) < 0$, 当 $k > 1 - \frac{1}{\ln \ln 2}$ 时,

$f'(k) > 0$, 故 $k = 1 - \frac{1}{\ln \ln 2}$ 为函数 $f(k)$ 的最小值点, 即当 $k = 1 - \frac{1}{\ln \ln 2}$ 时所给反常积分

取得最小值.

例 4. 利用递推公式计算反常积分 $I_n = \int_0^{+\infty} x^n e^{-x} dx \quad (n \in \mathbf{N})$.

$$\text{解 } I_0 = \int_0^{+\infty} e^{-x} dx = [-e^{-x}]_0^{+\infty} = 1.$$

当 $n \geq 1$ 时,

$$I_n = - \int_0^{+\infty} x^n d(e^{-x}) = -[x^n e^{-x}]_0^{+\infty} + n \int_0^{+\infty} x^{n-1} e^{-x} dx = n I_{n-1},$$

故有 $I_n = n!$.

例 5. 计算反常积分 $\int_0^1 \ln x dx$.

$$\text{解 } \int \ln x dx = x \ln x - \int x \cdot \frac{1}{x} dx = x \ln x - x + C,$$

因此

$$\int_0^1 \ln x dx = [x \ln x - x]_0^1 = -1 - \lim_{x \rightarrow 0^+} (x \ln x - x) = -1.$$

* 习题 5-5 反常积分的审敛法 Γ 函数

例 1. 判定下列反常积分的收敛性:

$$(1) \int_0^{+\infty} \frac{x^2}{x^4 + x^2 + 1} dx; \quad (2) \int_1^{+\infty} \frac{dx}{x \sqrt[3]{x^2 + 1}};$$

$$(3) \int_1^{+\infty} \sin \frac{1}{x^2} dx;$$

$$(4) \int_0^{+\infty} \frac{dx}{1+x|\sin x|};$$

$$(5) \int_1^{+\infty} \frac{x \arctan x}{1+x^3} dx;$$

$$(6) \int_1^2 \frac{dx}{(\ln x)^3};$$

$$(7) \int_0^1 \frac{x^4 dx}{\sqrt{1-x^4}};$$

$$(8) \int_1^2 \frac{dx}{\sqrt[3]{x^2-3x+2}}.$$

解 (1) 由于 $\lim_{x \rightarrow +\infty} x^2 \cdot \frac{x^2}{x^4+x^2+1} = 1$, 因此 $\int_0^{+\infty} \frac{x^2}{x^4+x^2+1} dx$ 收敛.

(2) 由于 $\lim_{x \rightarrow +\infty} x^{\frac{5}{3}} \cdot \frac{1}{x\sqrt[3]{x^2+1}} = 1$, 因此 $\int_1^{+\infty} \frac{dx}{x\sqrt[3]{x^2+1}}$ 收敛.

(3) 由于 $\lim_{x \rightarrow +\infty} x^2 \cdot \sin \frac{1}{x^2} = 1$, 因此 $\int_1^{+\infty} \sin \frac{1}{x^2} dx$ 收敛.

(4) 由于当 $x \geq 0$ 时, $\frac{1}{1+x|\sin x|} \geq \frac{1}{1+x}$, 且 $\int_0^{+\infty} \frac{dx}{1+x}$ 发散, 因此 $\int_0^{+\infty} \frac{dx}{1+x|\sin x|}$ 发散.

(5) 由于 $\lim_{x \rightarrow +\infty} x^2 \cdot \frac{x \arctan x}{1+x^3} = \frac{\pi}{2}$, 因此 $\int_1^{+\infty} \frac{x \arctan x}{1+x^3} dx$ 收敛.

(6) $x=1$ 是被积函数的瑕点. 由于 $\lim_{x \rightarrow 1^+} (x-1) \cdot \frac{1}{(\ln x)^3} = +\infty$, 因此 $\int_1^2 \frac{dx}{(\ln x)^3}$ 发散.

(7) $x=1$ 是被积函数的瑕点. 由于 $\lim_{x \rightarrow 1^-} (1-x)^{\frac{1}{2}} \cdot \frac{x^4}{\sqrt{1-x^4}} = \frac{1}{2}$, 因此 $\int_0^1 \frac{x^4 dx}{\sqrt{1-x^4}}$ 收敛.

(8) 被积函数有两个瑕点: $x=1, x=2$. 由于 $\lim_{x \rightarrow 1^+} (x-1)^{\frac{1}{3}} \cdot \frac{1}{\sqrt[3]{x^2-3x+2}} = -1$, 因此 $\int_1^{1.5} \frac{dx}{\sqrt[3]{x^2-3x+2}}$ 收敛; 又因为 $\lim_{x \rightarrow 2^-} (x-2)^{\frac{1}{3}} \cdot \frac{1}{\sqrt[3]{x^2-3x+2}} = 1$, 因此 $\int_{1.5}^2 \frac{dx}{\sqrt[3]{x^2-3x+2}}$ 收敛, 故 $\int_1^2 \frac{dx}{\sqrt[3]{x^2-3x+2}}$ 收敛.

2. 设反常积分 $\int_1^{+\infty} f^2(x) dx$ 收敛, 证明反常积分 $\int_1^{+\infty} \frac{f(x)}{x} dx$ 绝对收敛.

解 因为 $\left| \frac{f(x)}{x} \right| \leq \frac{f^2(x) + \frac{1}{x^2}}{2}$, 由于 $\int_1^{+\infty} f^2(x) dx$ 收敛, $\int_1^{+\infty} \frac{1}{x^2} dx$ 也收敛, 因此

$\int_1^{+\infty} \left| \frac{f(x)}{x} \right| dx$ 收敛, 即 $\int_1^{+\infty} \frac{f(x)}{x} dx$ 绝对收敛.

3. 用 Γ 函数表示下列积分, 并指出这些积分的收敛范围:

$$(1) \int_0^{+\infty} e^{-x^n} dx \quad (n > 0); \quad (2) \int_0^1 \left(\ln \frac{1}{x} \right)^p dx;$$

$$(3) \int_0^{+\infty} x^m e^{-x^n} dx \quad (n \neq 0).$$

解 (1) 令 $u = x^n$, 即 $x = u^{\frac{1}{n}}$,

$$\int_0^{+\infty} e^{-x^n} dx = \frac{1}{n} \int_0^{+\infty} e^{-u} u^{\frac{1}{n}-1} du = \frac{1}{n} \Gamma\left(\frac{1}{n}\right),$$

在 $n > 0$ 时都收敛.

(2) 令 $u = \ln \frac{1}{x}$, 即 $x = e^{-u}$,

$$\int_0^1 \left(\ln \frac{1}{x} \right)^p dx = \int_{+\infty}^0 -u^p e^{-u} du = \int_0^{+\infty} u^p e^{-u} du = \Gamma(p+1),$$

当 $p > -1$ 时收敛.

(3) 令 $u = x^n$, 即 $x = u^{\frac{1}{n}}$.

$$\text{当 } n > 0 \text{ 时, } \int_0^{+\infty} x^m e^{-x^n} dx = \int_0^{+\infty} \frac{1}{n} u^{\frac{m+1}{n}-1} e^{-u} du = \frac{1}{n} \Gamma\left(\frac{m+1}{n}\right),$$

$$\text{当 } n < 0 \text{ 时, } \int_0^{+\infty} x^m e^{-x^n} dx = \int_{+\infty}^0 \frac{1}{n} u^{\frac{m+1}{n}-1} e^{-u} du = -\frac{1}{n} \Gamma\left(\frac{m+1}{n}\right),$$

故 $\int_0^{+\infty} x^m e^{-x^n} dx = \frac{1}{|n|} \Gamma\left(\frac{m+1}{n}\right)$, 当 $\frac{m+1}{n} > 0$ 时收敛.

4. 证明 $\Gamma\left(\frac{2k+1}{2}\right) = \frac{1 \cdot 3 \cdot 5 \cdot \dots \cdot (2k-1) \sqrt{\pi}}{2^k}$, 其中 $k \in \mathbf{N}_+$.

$$\begin{aligned} \text{证 } \Gamma\left(\frac{2k+1}{2}\right) &= \frac{2k-1}{2} \Gamma\left(\frac{2k-1}{2}\right) = \frac{2k-1}{2} \cdot \frac{2k-3}{2} \Gamma\left(\frac{2k-3}{2}\right) \\ &= \frac{2k-1}{2} \cdot \frac{2k-3}{2} \cdot \dots \cdot \frac{1}{2} \Gamma\left(\frac{1}{2}\right) = \frac{1 \cdot 3 \cdot 5 \cdot \dots \cdot (2k-1)}{2^k} \sqrt{\pi}. \end{aligned}$$

5. 证明以下各式 (其中 $n \in \mathbf{N}_+$):

$$(1) 2 \cdot 4 \cdot 6 \cdot \dots \cdot (2n) = 2^n \Gamma(n+1);$$

$$(2) 1 \cdot 3 \cdot 5 \cdot \dots \cdot (2n-1) = \frac{\Gamma(2n)}{2^{n-1} \Gamma(n)};$$

$$(3) \sqrt{\pi} \Gamma(2n) = 2^{2n-1} \Gamma(n) \Gamma\left(n + \frac{1}{2}\right) \text{ (勒让德 (Legendre) 倍量公式)}.$$

证 (1) $2 \cdot 4 \cdot 6 \cdot \cdots \cdot (2n) = 2^n n! = 2^n \Gamma(n+1)$.

$$(2) 1 \cdot 3 \cdot 5 \cdot \cdots \cdot (2n-1) = \frac{(2n-1)!}{2 \cdot 4 \cdot 6 \cdot \cdots \cdot (2n-2)} = \frac{\Gamma(2n)}{2^{n-1}(n-1)!} = \frac{\Gamma(2n)}{2^{n-1}\Gamma(n)}.$$

(3) 因为 $\sqrt{\pi}\Gamma(2n) = (2n-1)! \sqrt{\pi}$,

$$\begin{aligned} \Gamma(n)\Gamma\left(n+\frac{1}{2}\right) &= (n-1)! \frac{1 \cdot 3 \cdot 5 \cdot \cdots \cdot (2n-1) \sqrt{\pi}}{2^n} \\ &= \frac{2 \cdot 4 \cdot 6 \cdot \cdots \cdot (2n-2)}{2^{n-1}} \cdot \frac{1 \cdot 3 \cdot 5 \cdot \cdots \cdot (2n-1) \sqrt{\pi}}{2^n} \\ &= \frac{(2n-1)!}{2^{2n-1}} \sqrt{\pi}, \end{aligned}$$

因此结论成立.

总 习 题 五

1. 填空:

(1) 函数 $f(x)$ 在 $[a, b]$ 上有界是 $f(x)$ 在 $[a, b]$ 上可积的 _____ 条件, 而 $f(x)$ 在 $[a, b]$ 上连续是 $f(x)$ 在 $[a, b]$ 上可积的 _____ 条件;

(2) 对 $[a, +\infty)$ 上非负、连续的函数 $f(x)$, 积分上限的函数 $\int_a^x f(t) dt$ 在 $[a, +\infty)$ 上有界是反常积分 $\int_a^{+\infty} f(x) dx$ 收敛的 _____ 条件;

* (3) 绝对收敛的反常积分 $\int_a^{+\infty} f(x) dx$ 一定 _____;

(4) 函数 $f(x)$ 在 $[a, b]$ 上有定义且 $|f(x)|$ 在 $[a, b]$ 上可积, 此时积分 $\int_a^b f(x) dx$ _____ 存在;

(5) 设函数 $f(x)$ 连续, 则 $\frac{d}{dx} \int_0^x t f(t^2 - x^2) dt =$ _____.

解 (1) 必要, 充分. (2) 充分必要. (3) 收敛.

(4) 不一定. 例如 $f(x) = \begin{cases} 1, & x \text{ 为有理数,} \\ -1, & x \text{ 为无理数,} \end{cases}$ 则 $|f(x)| = 1$ 在 $[a, b]$ 上可积, 而 $\int_a^b f(x) dx$ 不存在.

(5) $xf(-x^2)$. 作换元 $u = t^2 - x^2$, 则

$$\begin{aligned} \int_0^x t f(t^2 - x^2) dt &= \frac{1}{2} \int_0^x f(t^2 - x^2) d(t^2 - x^2) = \frac{1}{2} \int_{-x^2}^0 f(u) du \\ &= -\frac{1}{2} \int_0^{-x^2} f(u) du, \end{aligned}$$

因此

$$\frac{d}{dx} \int_0^x t f(t^2 - x^2) dt = -\frac{1}{2} f(-x^2) \cdot (-2x) = x f(-x^2).$$

2. 以下两题中给出了四个结论, 从中选出一个正确的结论:

(1) 设 $I = \int_0^1 \frac{x^4}{\sqrt{1+x}} dx$, 则估计 I 值的大致范围为();

(A) $0 \leq I \leq \frac{\sqrt{2}}{10}$ (B) $\frac{\sqrt{2}}{10} \leq I \leq \frac{1}{5}$

(C) $\frac{1}{5} < I < 1$ (D) $I \geq 1$

(2) 设 $F(x)$ 是连续函数 $f(x)$ 的一个原函数, 则必有().

(A) $F(x)$ 是偶函数 $\Leftrightarrow f(x)$ 是奇函数

(B) $F(x)$ 是奇函数 $\Leftrightarrow f(x)$ 是偶函数

(C) $F(x)$ 是周期函数 $\Leftrightarrow f(x)$ 是周期函数

(D) $F(x)$ 是单调函数 $\Leftrightarrow f(x)$ 是单调函数

解 (1) 当 $0 \leq x \leq 1$ 时, $\frac{1}{\sqrt{2}} x^4 \leq \frac{x^4}{\sqrt{1+x}} \leq x^4$, 因此

$$\frac{\sqrt{2}}{10} = \int_0^1 \frac{1}{\sqrt{2}} x^4 dx \leq \int_0^1 \frac{x^4}{\sqrt{1+x}} dx \leq \int_0^1 x^4 dx = \frac{1}{5},$$

故选(B).

(2) 记 $G(x) = \int_0^x f(t) dt$, 则 $G(x)$ 是 $f(x)$ 的一个原函数, 且

$G(x)$ 是奇(偶)函数 $\Leftrightarrow f(x)$ 是偶(奇)函数,

又 $F(x) = G(x) + C$, 其中 C 是一常数, 而常数是偶函数, 故由奇、偶函数的性质知应选(A).

取周期函数 $f(x) = \cos x + 1$, 则 $F(x) = \sin x + x + C$ 不是周期函数, 故(C)不成立; 取单调增加函数 $f(x) = 2x, x \in \mathbf{R}$, 则 $F(x) = x^2 + C$ 在 $\mathbf{R}$ 上不是单调函数, 故(D)不成立.

3. 回答下列问题:

(1) 设函数 $f(x)$ 及 $g(x)$ 在区间 $[a, b]$ 上连续, 且 $f(x) \geq g(x)$, 则 $\int_a^b [f(x) - g(x)] dx$ 在几何上表示什么?

(2) 设函数 $f(x)$ 在区间 $[a, b]$ 上连续, 且 $f(x) \geq 0$, 则 $\int_a^b \pi f^2(x) dx$ 在几何上表示什么?

(3) 如果在时刻 t 以 $\varphi(t)$ 的流量(单位时间内流过的流体的体积或质量)向一水池注水, 那么 $\int_{t_1}^{t_2} \varphi(t) dt$ 表示什么?

(4) 如果某国人口增长的速率为 $u(t)$, 那么 $\int_{T_1}^{T_2} u(t) dt$ 表示什么?

(5) 如果一公司经营某种产品的边际利润函数为 $P'(x)$, 那么 $\int_{1000}^{2000} P'(x) dx$ 表示什么?

解 (1) $\int_a^b [f(x) - g(x)] dx$ 表示由曲线 $y = f(x)$, $y = g(x)$ 以及直线 $x = a, x = b$ 所围成的图形的

面积.

(2) $\int_a^b \pi f^2(x) dx$ 表示 xOy 面上, 由曲线 $y=f(x)$, $x=a$, $x=b$ 以及 x 轴所围成的图形绕 x 轴旋转一周而得到的旋转体的体积.

(3) $\int_{t_1}^{t_2} \varphi(t) dt$ 表示在时间段 $[t_1, t_2]$ 内向水池注入的水的总量.

(4) $\int_{T_1}^{T_2} u(t) dt$ 表示该国在 $[T_1, T_2]$ 时间段内增加的人口总量.

(5) $\int_{1000}^{2000} P'(x) dx$ 表示从经营第 1000 个产品起一直到第 2000 个产品的利润总量.

 4. 利用定积分的定义计算下列极限:

$$(1) \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n \sqrt{1 + \frac{i}{n}}; \quad (2) \lim_{n \rightarrow \infty} \frac{1^p + 2^p + \cdots + n^p}{n^{p+1}} \quad (p > 0).$$

解 (1) $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n \sqrt{1 + \frac{i}{n}} = \int_0^1 \sqrt{1+x} dx = \left[\frac{2}{3} (1+x)^{\frac{3}{2}} \right]_0^1 = \frac{2}{3} (2\sqrt{2} - 1).$

$$(2) \lim_{n \rightarrow \infty} \frac{1^p + 2^p + \cdots + n^p}{n^{p+1}} = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n \left(\frac{i}{n} \right)^p = \int_0^1 x^p dx = \frac{1}{p+1}.$$

 5. 求下列极限:

$$(1) \lim_{x \rightarrow a} \frac{x}{x-a} \int_a^x f(t) dt, \text{ 其中 } f(x) \text{ 连续}; \quad (2) \lim_{x \rightarrow +\infty} \frac{\int_0^x (\arctan t)^2 dt}{\sqrt{x^2 + 1}}.$$

解 (1) 记 $F(x) = x \int_a^x f(t) dt$,

$$\lim_{x \rightarrow a} \frac{x}{x-a} \int_a^x f(t) dt = \lim_{x \rightarrow a} \frac{F(x) - F(a)}{x-a} = F'(a) = af(a).$$

(2) 先证明所求极限为未定式 $\frac{\infty}{\infty}$. 由于当 $x > \tan 1$ 时, $\arctan x > 1$, 记 $c = \int_0^{\tan 1} (\arctan t)^2 dt$, 则当

$x > \tan 1$ 时, 有

$$\int_0^x (\arctan t)^2 dt = c + \int_{\tan 1}^x (\arctan t)^2 dt > c + \int_{\tan 1}^x dt = c + x - \tan 1;$$

故有 $\lim_{x \rightarrow +\infty} \int_0^x (\arctan t)^2 dt = +\infty$, 从而利用洛必达法则有

$$\lim_{x \rightarrow +\infty} \frac{\int_0^x (\arctan t)^2 dt}{\sqrt{x^2 + 1}} = \lim_{x \rightarrow +\infty} \frac{(\arctan x)^2}{x} = \frac{\pi^2}{4}.$$

 6. 下列计算是否正确, 试说明理由:

$$(1) \int_{-1}^1 \frac{dx}{1+x^2} = - \int_{-1}^1 \frac{d\left(\frac{1}{x}\right)}{1+\left(\frac{1}{x}\right)^2} = \left[-\arctan \frac{1}{x} \right]_{-1}^1 = -\frac{\pi}{2};$$

(2) 因为

$$\int_{-1}^1 \frac{dx}{x^2+x+1} \stackrel{x=\frac{1}{t}}{=} \int_{-1}^1 \frac{dt}{t^2+t+1},$$

所以

$$\int_{-1}^1 \frac{dx}{x^2+x+1} = 0.$$

解 (1) 不对. 因为 $u = \frac{1}{x}$ 在 $[-1, 1]$ 上有间断点 $x=0$, 不符合换元法的要求. 而由习题 5-1 的第 12 题第(1)小题可知该积分一定为正, 因此该积分计算不对. 事实上,

$$\int_{-1}^1 \frac{dx}{1+x^2} = [\arctan x]_{-1}^1 = \frac{\pi}{2}.$$

(2) 不对. 原因与(1)相同. 事实上,

$$\begin{aligned} \int_{-1}^1 \frac{dx}{x^2+x+1} &= \int_{-1}^1 \frac{1}{\left(x+\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} d\left(x+\frac{1}{2}\right) \\ &= \left[\frac{2}{\sqrt{3}} \arctan \frac{2x+1}{\sqrt{3}} \right]_{-1}^1 = \frac{\pi}{\sqrt{3}}. \end{aligned}$$

7. 设 $x>0$, 证明 $\int_0^x \frac{1}{1+t^2} dt + \int_0^{\frac{1}{x}} \frac{1}{1+t^2} dt = \frac{\pi}{2}$.

证 记 $f(x) = \int_0^x \frac{1}{1+t^2} dt + \int_0^{\frac{1}{x}} \frac{1}{1+t^2} dt$, 则当 $x>0$ 时, 有

$$f'(x) = \frac{1}{1+x^2} + \frac{1}{1+\frac{1}{x^2}} \cdot \left(-\frac{1}{x^2}\right) = 0,$$

由拉格朗日中值定理的推论, 得

$$f(x) \equiv C \quad (x>0).$$

而 $f(1) = \int_0^1 \frac{1}{1+t^2} dt + \int_0^1 \frac{1}{1+t^2} dt = \frac{\pi}{2}$, 故 $C = \frac{\pi}{2}$, 从而结论成立.

8. 设 $p>0$, 证明

$$\frac{p}{1+p} < \int_0^1 \frac{dx}{1+x^p} < 1.$$

证 由于当 $p>0, 0<x<1$ 时, $0 < \frac{1}{1+x^p} < 1$, 因此有 $\int_0^1 \frac{dx}{1+x^p} < 1$. 又

$$1 - \int_0^1 \frac{dx}{1+x^p} = \int_0^1 \frac{x^p dx}{1+x^p} < \int_0^1 x^p dx = \frac{1}{1+p},$$

故有 $\int_0^1 \frac{dx}{1+x^p} > \frac{p}{1+p}$, 原题得证.

9. 设 $f(x), g(x)$ 在区间 $[a, b]$ 上均连续, 证明:

$$(1) \left[\int_a^b f(x)g(x) dx \right]^2 \leq \int_a^b f^2(x) dx \cdot \int_a^b g^2(x) dx \quad (\text{柯西-施瓦茨不等式});$$

$$(2) \left\{ \int_a^b [f(x) + g(x)]^2 dx \right\}^{\frac{1}{2}} \leq \left[\int_a^b f^2(x) dx \right]^{\frac{1}{2}} + \left[\int_a^b g^2(x) dx \right]^{\frac{1}{2}} \quad (\text{闵可夫斯基不等式}).$$

证 (1) 对任意实数 λ , 有 $\int_a^b [f(x) + \lambda g(x)]^2 dx \geq 0$, 即

$$\int_a^b f^2(x) dx + 2\lambda \int_a^b f(x)g(x) dx + \lambda^2 \int_a^b g^2(x) dx \geq 0,$$

上式左边是一个关于 λ 的二次三项式, 它非负的条件是其系数判别式非正, 即有

$$4 \left[\int_a^b f(x)g(x) dx \right]^2 - 4 \int_a^b f^2(x) dx \cdot \int_a^b g^2(x) dx \leq 0,$$

从而本题得证.

$$\begin{aligned} (2) \int_a^b [f(x) + g(x)]^2 dx &= \int_a^b [f^2(x) + 2f(x)g(x) + g^2(x)] dx \\ &= \int_a^b f^2(x) dx + 2 \int_a^b f(x)g(x) dx + \int_a^b g^2(x) dx \\ &\leq \int_a^b f^2(x) dx + 2 \left[\int_a^b f^2(x) dx \int_a^b g^2(x) dx \right]^{\frac{1}{2}} + \int_a^b g^2(x) dx \\ &= \left\{ \left[\int_a^b f^2(x) dx \right]^{\frac{1}{2}} + \left[\int_a^b g^2(x) dx \right]^{\frac{1}{2}} \right\}^2, \end{aligned}$$

从而本题得证.

 10. 设 $f(x)$ 在区间 $[a, b]$ 上连续, 且 $f(x) > 0$. 证明

$$\int_a^b f(x) dx \cdot \int_a^b \frac{1}{f(x)} dx \geq (b-a)^2.$$

证 根据上一题所证的柯西-施瓦茨不等式, 有

$$\left[\int_a^b \sqrt{f(x)} \cdot \frac{1}{\sqrt{f(x)}} dx \right]^2 \leq \int_a^b [\sqrt{f(x)}]^2 dx \cdot \int_a^b \left[\frac{1}{\sqrt{f(x)}} \right]^2 dx,$$

即得

$$\int_a^b f(x) dx \cdot \int_a^b \frac{1}{f(x)} dx \geq (b-a)^2.$$

 11. 计算下列积分:

$$(1) \int_0^{\frac{\pi}{2}} \frac{x + \sin x}{1 + \cos x} dx;$$

$$(2) \int_0^{\frac{\pi}{4}} \ln(1 + \tan x) dx;$$

$$(3) \int_0^a \frac{dx}{x + \sqrt{a^2 - x^2}} \quad (a > 0);$$

$$(4) \int_0^{\frac{\pi}{2}} \sqrt{1 - \sin 2x} dx;$$

$$(5) \int_0^{\frac{\pi}{2}} \frac{dx}{1 + \cos^2 x};$$

$$(6) \int_0^{\pi} x \sqrt{\cos^2 x - \cos^4 x} dx;$$

$$(7) \int_0^{\pi} x^2 |\cos x| dx;$$

$$(8) \int_0^{+\infty} \frac{dx}{e^{x+1} + e^{3-x}};$$

$$(9) \int_{\frac{1}{2}}^{\frac{3}{2}} \frac{dx}{\sqrt{|x^2 - x|}};$$

$$(10) \int_0^x \max\{t^3, t^2, 1\} dt.$$

解 (1) $\int_0^{\frac{\pi}{2}} \frac{x + \sin x}{1 + \cos x} dx = \int_0^{\frac{\pi}{2}} \frac{x}{1 + \cos x} dx + \int_0^{\frac{\pi}{2}} \frac{\sin x}{1 + \cos x} dx$

$$\begin{aligned}
&= \int_0^{\frac{\pi}{2}} \frac{x}{2} \sec^2 \frac{x}{2} dx - \int_0^{\frac{\pi}{2}} \frac{1}{1+\cos x} d(1+\cos x) \\
&= \left[x \tan \frac{x}{2} \right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \tan \frac{x}{2} dx - \left[\ln(1+\cos x) \right]_0^{\frac{\pi}{2}} \\
&= \frac{\pi}{2} + \left[2 \ln \cos \frac{x}{2} \right]_0^{\frac{\pi}{2}} + \ln 2 = \frac{\pi}{2}.
\end{aligned}$$

$$\begin{aligned}
(2) \int_0^{\frac{\pi}{4}} \ln(1+\tan x) dx &= \int_0^{\frac{\pi}{4}} \ln \frac{\cos x + \sin x}{\cos x} dx \\
&= \int_0^{\frac{\pi}{4}} \ln(\cos x + \sin x) dx - \int_0^{\frac{\pi}{4}} \ln \cos x dx,
\end{aligned}$$

而

$$\begin{aligned}
\int_0^{\frac{\pi}{4}} \ln(\cos x + \sin x) dx &= \int_0^{\frac{\pi}{4}} \ln \left[\sqrt{2} \cos \left(\frac{\pi}{4} - x \right) \right] dx \\
&\stackrel{x = \frac{\pi}{4} - u}{=} - \int_{\frac{\pi}{4}}^0 (\ln \sqrt{2} + \ln \cos u) du \\
&= \frac{\pi \ln 2}{8} + \int_0^{\frac{\pi}{4}} \ln \cos x dx,
\end{aligned}$$

故

$$\begin{aligned}
&\int_0^{\frac{\pi}{4}} \ln(1+\tan x) dx = \frac{\pi \ln 2}{8}. \\
(3) \int_0^a \frac{dx}{x + \sqrt{a^2 - x^2}} &\stackrel{x = a \sin u}{=} \int_0^{\frac{\pi}{2}} \frac{\cos u du}{\sin u + \cos u} = \int_0^{\frac{\pi}{2}} \frac{\sin u du}{\cos u + \sin u} \\
&= \frac{1}{2} \left(\int_0^{\frac{\pi}{2}} \frac{\cos u du}{\sin u + \cos u} + \int_0^{\frac{\pi}{2}} \frac{\sin u du}{\cos u + \sin u} \right) \\
&= \frac{1}{2} \int_0^{\frac{\pi}{2}} du = \frac{\pi}{4}.
\end{aligned}$$

$$\begin{aligned}
(4) \int_0^{\frac{\pi}{2}} \sqrt{1 - \sin 2x} dx &= \int_0^{\frac{\pi}{2}} \sqrt{\sin^2 x + \cos^2 x - 2 \sin x \cos x} dx \\
&= \int_0^{\frac{\pi}{2}} |\sin x - \cos x| dx \\
&= \int_0^{\frac{\pi}{4}} (\cos x - \sin x) dx + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (\sin x - \cos x) dx \\
&= \left[\sin x + \cos x \right]_0^{\frac{\pi}{4}} + \left[-\cos x - \sin x \right]_{\frac{\pi}{4}}^{\frac{\pi}{2}} \\
&= 2(\sqrt{2} - 1).
\end{aligned}$$

(5) 注意到 $\lim_{x \rightarrow \frac{\pi}{2}^-} \arctan \frac{\tan x}{\sqrt{2}} = \frac{\pi}{2}$, 因此有

$$\int_0^{\frac{\pi}{2}} \frac{dx}{1 + \cos^2 x} = \int_0^{\frac{\pi}{2}} \frac{\sec^2 x dx}{\sec^2 x + 1} = \int_0^{\frac{\pi}{2}} \frac{d(\tan x)}{\tan^2 x + 2}$$

$$= \left[\frac{1}{\sqrt{2}} \arctan \frac{\tan x}{\sqrt{2}} \right]_0^{\frac{\pi}{2}} = \frac{\pi}{2\sqrt{2}}.$$

$$\begin{aligned} (6) \int_0^{\pi} x \sqrt{\cos^2 x - \cos^4 x} dx &= \int_0^{\pi} x |\cos x| \sin x dx \\ &= \frac{\pi}{2} \int_0^{\pi} |\cos x| \sin x dx \\ &= \frac{\pi}{2} \left[\int_0^{\frac{\pi}{2}} \cos x \sin x dx - \int_{\frac{\pi}{2}}^{\pi} \cos x \sin x dx \right] \\ &= \frac{\pi}{2} \left[\frac{1}{2} \sin^2 x \right]_0^{\frac{\pi}{2}} - \frac{\pi}{2} \left[\frac{1}{2} \sin^2 x \right]_{\frac{\pi}{2}}^{\pi} = \frac{\pi}{2}. \end{aligned}$$

$$\begin{aligned} (7) \int_0^{\pi} x^2 |\cos x| dx &= \int_0^{\frac{\pi}{2}} x^2 \cos x dx - \int_{\frac{\pi}{2}}^{\pi} x^2 \cos x dx \\ &= [x^2 \sin x + 2x \cos x - 2 \sin x]_0^{\frac{\pi}{2}} - [x^2 \sin x + 2x \cos x - 2 \sin x]_{\frac{\pi}{2}}^{\pi} \\ &= \frac{\pi^2}{2} + 2\pi - 4. \end{aligned}$$

$$\begin{aligned} (8) \int_0^{+\infty} \frac{dx}{e^{x+1} + e^{3-x}} &= \frac{1}{e^2} \int_0^{+\infty} \frac{d(e^{x-1})}{e^{2x-2} + 1} = \frac{1}{e^2} [\arctan(e^{x-1})]_0^{+\infty} \\ &= \frac{1}{e^2} \left(\frac{\pi}{2} - \arctan \frac{1}{e} \right). \end{aligned}$$

$$\begin{aligned} (9) \int_{\frac{1}{2}}^1 \frac{dx}{\sqrt{|x^2 - x|}} &= \int_{\frac{1}{2}}^1 \frac{dx}{\sqrt{x - x^2}} = \int_{\frac{1}{2}}^1 \frac{d(2x - 1)}{\sqrt{1 - (2x - 1)^2}}; \\ &= [\arcsin(2x - 1)]_{\frac{1}{2}}^1 = \frac{\pi}{2}; \\ \int_1^{\frac{3}{2}} \frac{dx}{\sqrt{|x^2 - x|}} &= \int_1^{\frac{3}{2}} \frac{dx}{\sqrt{x^2 - x}} = \int_1^{\frac{3}{2}} \frac{d(2x - 1)}{\sqrt{(2x - 1)^2 - 1}} \\ &= [\ln(2x - 1 + \sqrt{(2x - 1)^2 - 1})]_1^{\frac{3}{2}} = \ln(2 + \sqrt{3}), \end{aligned}$$

因此

$$\begin{aligned} \int_{\frac{1}{2}}^{\frac{3}{2}} \frac{dx}{\sqrt{|x^2 - x|}} &= \int_{\frac{1}{2}}^1 \frac{dx}{\sqrt{|x^2 - x|}} + \int_1^{\frac{3}{2}} \frac{dx}{\sqrt{|x^2 - x|}} \\ &= \frac{\pi}{2} + \ln(2 + \sqrt{3}). \end{aligned}$$

(10) 当 $x < -1$ 时,

$$\int_0^x \max\{t^3, t^2, 1\} dt = \int_0^{-1} dt + \int_{-1}^x t^2 dt = \frac{1}{3}x^3 - \frac{2}{3};$$

当 $-1 \leq x \leq 1$ 时,

$$\int_0^x \max\{t^3, t^2, 1\} dt = \int_0^x dt = x;$$

当 $x > 1$ 时,

$$\int_0^x \max\{t^3, t^2, 1\} dt = \int_0^1 dt + \int_1^x t^3 dt = \frac{1}{4}x^4 + \frac{3}{4}.$$

因此

$$\int_0^x \max\{t^3, t^2, 1\} dt = \begin{cases} \frac{1}{3}x^3 - \frac{2}{3}, & x < -1, \\ x, & -1 \leq x \leq 1, \\ \frac{1}{4}x^4 + \frac{3}{4}, & x > 1. \end{cases}$$

 12. 设 $f(x)$ 为连续函数, 证明

$$\int_0^x f(t)(x-t) dt = \int_0^x \left[\int_0^t f(u) du \right] dt.$$

$$\begin{aligned} \text{证} \quad \int_0^x \left[\int_0^t f(u) du \right] dt &= \left[t \int_0^t f(u) du \right]_0^x - \int_0^x tf(t) dt \\ &= x \int_0^x f(u) du - \int_0^x tf(t) dt \\ &= x \int_0^x f(t) dt - \int_0^x tf(t) dt = \int_0^x (x-t)f(t) dt. \end{aligned}$$

本题也可利用原函数性质来证明, 记等式左端的函数为 $F(x)$ 、右端的函数为 $G(x)$, 则

$$F'(x) = \left[x \int_0^x f(t) dt - \int_0^x tf(t) dt \right]' = \int_0^x f(t) dt,$$

$$G'(x) = \int_0^x f(u) du = \int_0^x f(t) dt,$$

即 $F(x), G(x)$ 都为函数 $\int_0^x f(t) dt$ 的原函数, 因此它们至多只差一个常数, 但由于 $F(0) = G(0) = 0$, 因此必有 $F(x) = G(x)$.

 13. 设 $f(x)$ 在区间 $[a, b]$ 上连续, 且 $f(x) > 0$,

$$F(x) = \int_a^x f(t) dt + \int_b^x \frac{dt}{f(t)}, x \in [a, b].$$

证明: (1) $F'(x) \geq 2$; (2) 方程 $F(x) = 0$ 在区间 (a, b) 内有且仅有一个根.

$$\text{证} \quad (1) \quad F'(x) = f(x) + \frac{1}{f(x)} \geq 2\sqrt{f(x) \cdot \frac{1}{f(x)}} = 2.$$

(2) $F(a) = \int_b^a \frac{dt}{f(t)} = -\int_a^b \frac{dt}{f(t)} < 0, f(b) = \int_a^b f(t) dt > 0$, 由闭区间上连续函数性质可知 $F(x)$ 在区间 (a, b) 内必有零点, 根据 (1) 可知函数 $F(x)$ 在区间 $[a, b]$ 上单调增加, 从而零点唯一, 即方程 $F(x) = 0$ 在区间 (a, b) 内有且仅有一个根.

 14. 求 $\int_0^2 f(x-1) dx$, 其中

$$f(x) = \begin{cases} \frac{1}{1+e^x}, & x < 0, \\ \frac{1}{1+x}, & x \geq 0. \end{cases}$$

$$\begin{aligned}
 \text{解} \quad \int_0^2 f(x-1) dx & \stackrel{x=u+1}{=} \int_{-1}^1 f(u) du = \int_{-1}^0 \frac{du}{1+e^u} + \int_0^1 \frac{du}{1+u} \\
 & = \int_{-1}^0 \frac{e^{-u} du}{1+e^{-u}} + [\ln(1+u)]_0^1 \\
 & = [-\ln(1+e^{-u})]_{-1}^0 + \ln 2 = \ln(1+e).
 \end{aligned}$$

 15. 设 $f'(x)$ 在 $[-\pi, \pi]$ 上连续, $a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx (n \in \mathbf{N})$. 证明: $\lim_{n \rightarrow \infty} a_n = 0$.

证 利用分部积分法.

$$\begin{aligned}
 a_n &= \frac{1}{n\pi} \int_{-\pi}^{\pi} f(x) d(\sin nx) \\
 &= \frac{1}{n\pi} \left[f(x) \sin nx \Big|_{-\pi}^{\pi} - \int_{-\pi}^{\pi} f'(x) \sin nx dx \right] \\
 &= \frac{-1}{n\pi} \int_{-\pi}^{\pi} f'(x) \sin nx dx.
 \end{aligned}$$

由于 $|f'(x) \sin nx| \leq |f'(x)|$, 而 $f'(x)$ 在 $[-\pi, \pi]$ 上连续, 故 $f'(x)$ 在 $[-\pi, \pi]$ 上有界, 即存在 $M > 0$, 使 $|f'(x)| \leq M (x \in [-\pi, \pi])$, 于是

$$\begin{aligned}
 |a_n| &= \frac{1}{n\pi} \left| \int_{-\pi}^{\pi} f'(x) \sin nx dx \right| \\
 &\leq \frac{1}{n\pi} \int_{-\pi}^{\pi} |f'(x) \sin nx| dx \leq \frac{1}{n\pi} \int_{-\pi}^{\pi} |f'(x)| dx \\
 &\leq \frac{M}{n\pi} \int_{-\pi}^{\pi} dx = \frac{M}{n\pi} \cdot 2\pi = \frac{2M}{n}.
 \end{aligned}$$

当 $n \rightarrow \infty$ 时, $\frac{2M}{n} \rightarrow 0$, 故 $|a_n| \rightarrow 0$. 从而得 $\lim_{n \rightarrow \infty} a_n = 0$.

 16. 设 $f(x)$ 在区间 $[a, b]$ 上连续, $g(x)$ 在区间 $[a, b]$ 上连续不变号. 证明至少存在一点 $\xi \in [a, b]$, 使下式成立:

$$\int_a^b f(x)g(x) dx = f(\xi) \int_a^b g(x) dx \quad (\text{积分第一中值定理}).$$

证 不妨设 $g(x) \geq 0$, 由定积分性质可知 $\int_a^b g(x) dx \geq 0$. 记 $f(x)$ 在 $[a, b]$ 上的最大值为 M 、最小值为 m , 则有

$$mg(x) \leq f(x)g(x) \leq Mg(x),$$

故有

$$\begin{aligned}
 m \int_a^b g(x) dx &= \int_a^b mg(x) dx \leq \int_a^b f(x)g(x) dx \\
 &\leq \int_a^b Mg(x) dx = M \int_a^b g(x) dx.
 \end{aligned}$$

当 $\int_a^b g(x) dx = 0$ 时, 由上述不等式可知 $\int_a^b f(x)g(x) dx = 0$, 故结论成立.

当 $\int_a^b g(x) dx > 0$ 时, 有

$$m \leq \frac{\int_a^b f(x)g(x)dx}{\int_a^b g(x)dx} \leq M,$$

由闭区间上连续函数性质, 知存在 $\xi \in [a, b]$, 使得

$$f(\xi) = \frac{\int_a^b f(x)g(x)dx}{\int_a^b g(x)dx},$$

从而结论成立.

 * 17. 证明: $\int_0^{+\infty} x^n e^{-x^2} dx = \frac{n-1}{2} \int_0^{+\infty} x^{n-2} e^{-x^2} dx$ ($n > 1$), 并用它证明:

$$\int_0^{+\infty} x^{2n+1} e^{-x^2} dx = \frac{1}{2} \Gamma(n+1) \quad (n \in \mathbf{N}).$$

证 当 $n > 1$ 时,

$$\begin{aligned} \int_0^{+\infty} x^n e^{-x^2} dx &= -\frac{1}{2} \int_0^{+\infty} x^{n-1} d(e^{-x^2}) \\ &= -\frac{1}{2} [x^{n-1} e^{-x^2}]_0^{+\infty} + \frac{n-1}{2} \int_0^{+\infty} x^{n-2} e^{-x^2} dx \\ &= \frac{n-1}{2} \int_0^{+\infty} x^{n-2} e^{-x^2} dx. \end{aligned}$$

记 $I_n = \int_0^{+\infty} x^{2n+1} e^{-x^2} dx$, 则

$$I_n = \int_0^{+\infty} x^{2n+1} e^{-x^2} dx = \frac{2n+1-1}{2} \int_0^{+\infty} x^{2n-1} e^{-x^2} dx = n \int_0^{+\infty} x^{2n-1} e^{-x^2} dx = n I_{n-1},$$

因此有

$$I_n = n! I_0 = n! \int_0^{+\infty} x e^{-x^2} dx = n! \left[-\frac{1}{2} e^{-x^2} \right]_0^{+\infty} = \frac{1}{2} n! = \frac{1}{2} \Gamma(n+1).$$

 * 18. 判断下列反常积分的收敛性:

$$\begin{aligned} (1) \int_0^{+\infty} \frac{\sin x}{\sqrt{x^3}} dx; & \quad (2) \int_2^{+\infty} \frac{dx}{x \cdot \sqrt[3]{x^2-3x+2}}; \\ (3) \int_2^{+\infty} \frac{\cos x}{\ln x} dx; & \quad (4) \int_0^{+\infty} \frac{dx}{\sqrt[3]{x^2(x-1)(x-2)}}. \end{aligned}$$

解 (1) $x=0$ 为被积函数 $f(x) = \frac{\sin x}{\sqrt{x^3}}$ 的瑕点, 而 $\lim_{x \rightarrow 0^+} x^{\frac{1}{2}} \cdot f(x) = 1$, 因此 $\int_0^1 f(x) dx$ 收敛; 又由于

$|f(x)| \leq \frac{1}{\sqrt{x^3}}$, 而 $\int_1^{+\infty} \frac{1}{\sqrt{x^3}} dx$ 收敛, 故 $\int_1^{+\infty} f(x) dx$ 收敛, 因此 $\int_0^{+\infty} \frac{\sin x}{\sqrt{x^3}} dx$ 收敛.

(2) $x=2$ 为被积函数 $f(x) = \frac{1}{x \cdot \sqrt[3]{x^2-3x+2}}$ 的瑕点, 而

$$\lim_{x \rightarrow 2^+} (x-2)^{\frac{1}{3}} \cdot f(x) = \frac{1}{2},$$

因此 $\int_2^3 f(x) dx$ 收敛; 又由于 $\lim_{x \rightarrow +\infty} x^{\frac{5}{3}} \cdot f(x) = 1$, 因此 $\int_3^{+\infty} \frac{dx}{x \cdot \sqrt[3]{x^2-3x+2}}$ 收敛, 故 $\int_2^{+\infty} \frac{dx}{x \cdot \sqrt[3]{x^2-3x+2}}$ 收敛.

$$\begin{aligned} (3) \int_2^{+\infty} \frac{\cos x}{\ln x} dx &= \int_2^{+\infty} \frac{1}{\ln x} d(\sin x) = \left[\frac{\sin x}{\ln x} \right]_2^{+\infty} + \int_2^{+\infty} \frac{\sin x}{x \ln^2 x} dx \\ &= \int_2^{+\infty} \frac{\sin x}{x \ln^2 x} dx - \frac{\sin 2}{\ln 2}, \end{aligned}$$

又由于 $\left| \frac{\sin x}{x \ln^2 x} \right| \leq \frac{1}{x \ln^2 x}$, 而 $\int_2^{+\infty} \frac{1}{x \ln^2 x} dx$ 收敛, 故 $\int_2^{+\infty} \left| \frac{\sin x}{x \ln^2 x} \right| dx$ 收敛, 即 $\int_2^{+\infty} \frac{\sin x}{x \ln^2 x} dx$ 绝对收敛, 因此 $\int_2^{+\infty} \frac{\cos x}{\ln x} dx$ 收敛.

$$(4) x=0, x=1, x=2 \text{ 为被积函数 } f(x) = \frac{1}{\sqrt[3]{x^2(x-1)(x-2)}} \text{ 的瑕点, } \lim_{x \rightarrow 0^+} x^{\frac{2}{3}} f(x) = \frac{1}{\sqrt[3]{2}},$$

$\lim_{x \rightarrow 1} (x-1)^{\frac{1}{3}} f(x) = -1, \lim_{x \rightarrow 2} f(x)(x-2)^{\frac{1}{3}} = \frac{\sqrt[3]{2}}{2}$, 故 $\int_0^3 f(x) dx$ 收敛; 又由于 $\lim_{x \rightarrow +\infty} x^{\frac{4}{3}} \cdot f(x) = 1$, 因此

$$\int_3^{+\infty} \frac{dx}{\sqrt[3]{x^2(x-1)(x-2)}} \text{ 收敛, 故 } \int_0^{+\infty} \frac{dx}{\sqrt[3]{x^2(x-1)(x-2)}} \text{ 收敛.}$$

例 19. 计算下列反常积分:

$$(1) \int_0^{\frac{\pi}{2}} \ln \sin x dx; \quad (2) \int_0^{+\infty} \frac{dx}{(1+x^2)(1+x^\alpha)} \quad (\alpha \geq 0).$$

解 (1) $x=0$ 为被积函数 $f(x) = \ln \sin x$ 的瑕点, 而

$$\lim_{x \rightarrow 0^+} \sqrt{x} \cdot f(x) = \lim_{x \rightarrow 0^+} \frac{\ln \sin x}{x^{-\frac{1}{2}}} = \lim_{x \rightarrow 0^+} \frac{\cot x}{-\frac{1}{2}x^{-\frac{3}{2}}} = \lim_{x \rightarrow 0^+} \frac{-2x^{\frac{3}{2}}}{\tan x} = 0,$$

故 $\int_0^{\frac{\pi}{2}} \ln \sin x dx$ 收敛.

$$\text{又 } \int_0^{\frac{\pi}{2}} \ln \sin x dx = \int_0^{\frac{\pi}{4}} \ln \sin x dx + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \ln \sin x dx, \text{ 而}$$

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \ln \sin x dx \stackrel{x = \frac{\pi}{2} - u}{=} \int_{\frac{\pi}{4}}^0 -\ln \cos u du = \int_0^{\frac{\pi}{4}} \ln \cos u du,$$

因此

$$\begin{aligned} \int_0^{\frac{\pi}{2}} \ln \sin x dx &= \int_0^{\frac{\pi}{4}} \ln \sin x dx + \int_0^{\frac{\pi}{4}} \ln \cos x dx \\ &= \int_0^{\frac{\pi}{4}} \ln(\sin x \cos x) dx = \int_0^{\frac{\pi}{4}} (\ln \sin 2x - \ln 2) dx \\ &= \int_0^{\frac{\pi}{4}} \ln \sin 2x dx - \int_0^{\frac{\pi}{4}} \ln 2 dx \\ &\stackrel{u=2x}{=} \frac{1}{2} \int_0^{\frac{\pi}{2}} \ln \sin u du - \frac{\pi}{4} \ln 2, \end{aligned}$$

故

$$\int_0^{\frac{\pi}{2}} \ln \sin x dx = -\frac{\pi}{2} \ln 2.$$

(2) 记被积函数为 $f(x) = \frac{1}{(1+x^2)(1+x^\alpha)}$, 则当 $\alpha = 0$ 时, $\lim_{x \rightarrow +\infty} x^2 \cdot f(x) = \frac{1}{2}$, 当 $\alpha > 0$ 时,

$\lim_{x \rightarrow +\infty} x^2 \cdot f(x) = 0$, 因此当 $\alpha \geq 0$ 时, $\int_0^{+\infty} \frac{dx}{(1+x^2)(1+x^\alpha)}$ 收敛.

令 $x = \frac{1}{t}$, 得到 $\int_0^{+\infty} \frac{dx}{(1+x^2)(1+x^\alpha)} = \int_{+\infty}^0 \frac{-t^\alpha dt}{(1+t^2)(1+t^\alpha)}$, 又

$$\int_{+\infty}^0 \frac{-t^\alpha dt}{(1+t^2)(1+t^\alpha)} = \int_0^{+\infty} \frac{x^\alpha dx}{(1+x^2)(1+x^\alpha)},$$

故

$$\begin{aligned} \int_0^{+\infty} \frac{dx}{(1+x^2)(1+x^\alpha)} &= \int_0^{+\infty} \frac{x^\alpha dx}{(1+x^2)(1+x^\alpha)} \\ &= \frac{1}{2} \left[\int_0^{+\infty} \frac{dx}{(1+x^2)(1+x^\alpha)} + \int_0^{+\infty} \frac{x^\alpha dx}{(1+x^2)(1+x^\alpha)} \right] \\ &= \frac{1}{2} \int_0^{+\infty} \frac{dx}{1+x^2} = \frac{1}{2} [\arctan x]_0^{+\infty} = \frac{\pi}{4}. \end{aligned}$$